```
$ ssh -X taller-locmarcaXXX@kabre…
$ echo $DISPLAY
localhost:10.0
$ ncview sst_costarica_2023.nc &
```

LOCMARCA 2026 · PRACTICAL GUIDE

ncview on Kabre with MobaXterm

Quick-look maps of a netCDF file, displayed on your own screen

WHY

Why not from the Jupyter notebook?

JupyterLab (OnDemand)

Runs in your browser: notebooks and a text terminal, but no way to open a separate window.

```
!ncview file.nc → Can't open display
```

MobaXterm (Windows)

An SSH terminal with a built-in **X server**: windows opened on Kabre appear on your screen.

```
ncview file.nc → a window opens
```

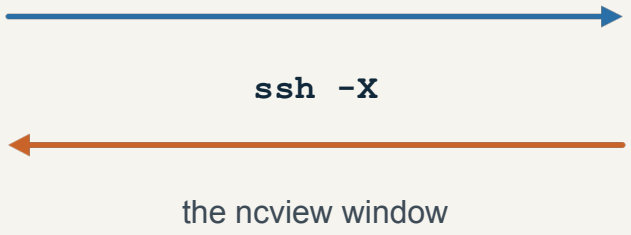
ncview is for a **quick look**: browse maps and time steps in seconds. Figures for your slides are made in Python.

HOW IT WORKS

ncview runs on Kabre, its window travels to you

Your laptop

MobaXterm: the SSH terminal where you type, and the X server that draws the windows



Kabre

ncview runs here and reads the file where it is stored: nothing to download

The **-X** option (X11 forwarding) is what carries the window back through the SSH connection.

STEP 1 · BEFORE THE WORKSHOP

Install MobaXterm

1

Download

mobaxterm.mobatek.net → Download → **Home Edition** (free)

2

Choose

Portable edition: unzip and run, no administrator rights needed. Or the Installer edition.

3

Launch

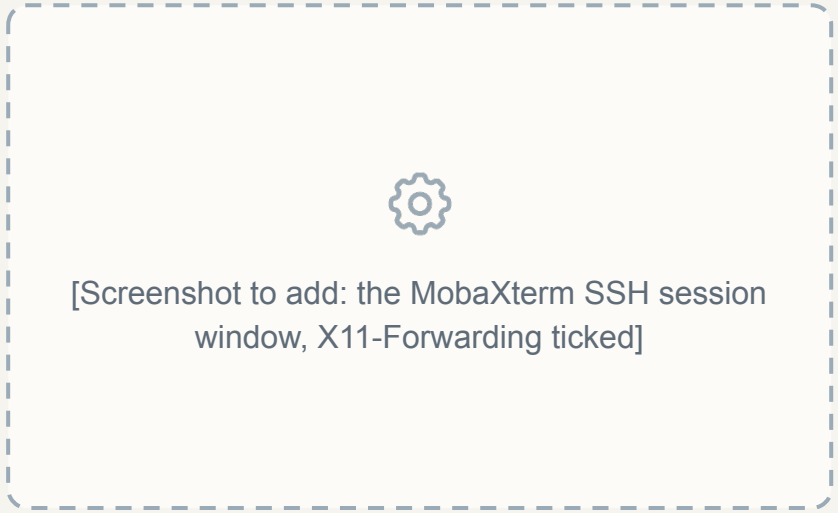
The X server starts on its own: the **X server** icon, top right, is in colour

⚠ Do it **at home, before Monday**: downloading on the workshop Wi-Fi with 20 people is slow.

STEP 2 · CONNECT

Create an SSH session to Kabre

1. Click **Session** (top left), then **SSH**
2. **Remote host**: kabre.cenat.ac.cr
3. Tick **Specify username**: taller-locmarcaXXX
4. Tab **Advanced SSH settings**: **X11-Forwarding** must be ticked
5. **OK**, then type your password: nothing appears as you type, that is normal



The session is saved in the left panel: next time, a double-click is enough.

STEP 2 · OR, WITH ONE COMMAND

The same connection, typed

In MobaXterm, click **Start local terminal**, then type:

```
ssh -X taller-locmarcaXXX@kabre.cenat.ac.cr
```

`ssh`	Open a secure connection
`-X`	Forward the windows (X11)
`user`	Your workshop account
`@host`	The Kabre login address

STEP 3 · CHECK

Two checks before launching

```
$ echo $DISPLAY
localhost:10.0
```

Any value: the window forwarding works. **Empty line**: reconnect with X11 forwarding (-X).

```
$ which ncview
…/psf2026/bin/ncview
```

ncview comes with the **psf2026** environment, loaded automatically at login. No answer: module load mamba/psf2026

STEP 4 · LAUNCH

Open a file with ncview

```
$ cd /data/gcambon/COMMONDATA/DATASETS_PSF/COSTARICA_DS
$ ls
$ ncview sst_costarica_2023.nc &
```

The &

Runs ncview in the background: your terminal stays free for other commands.

No copy

Open the shared files where they are: no need to copy them to your home.

Be patient

The first window can take a few seconds to appear over the network.

STEP 5 · EXPLORE

The ncview window in one slide

Button	What it does
> and <	Next / previous time step
>> and <<	Play the animation forward / backward
||	Pause
Colormap name	Click to change the colour palette
Inv C	Invert the colour palette
Range	Set the min / max of the colour scale
Mag	Enlarge the image (×2, ×3…)
Quit	Close ncview

Move the mouse

Shows the value and the coordinates under the cursor

Click on the map

Opens the time series at that point

Variable and dimension buttons

Choose the variable, the depth level, the season…

YOUR TURN · 10 MINUTES

Find the Papagayo cold event

1. Open **sst_costarica_2023.nc**, enlarge with **Mag**, play the year with **>>**
2. Stop in **February**: find the cold tongue west of the Gulf of Papagayo
3. **Click** near 87°W, 10°N and read the time series: when is it coldest?
4. Open **../GLOBAL_DS/WOA2009/temp_seas.cdf**: change the depth (Z) and the season (T)

Check: coldest around **20 February**, ≈ **297 K**. ncview shows the file's units: kelvin, so 297 K ≈ 24 °C.

TROUBLESHOOTING

When it does not work

You see	Cause	Fix
Can't open display	No X11 forwarding	Reconnect with -X, or tick X11-Forwarding
ncview: command not found	Environment not loaded	module load mamba/psf2026
No such file or directory	Wrong path or name	ls, then copy the exact name
Window very slow	Network: every redraw travels	Smaller file or region; close other windows
Nothing happens	Window hidden or still loading	Wait; look at the Windows task bar

NOT ON WINDOWS?

Mac and Linux

Mac

1. Install **XQuartz** (xquartz.org), then log out and back in
2. In the Terminal app:

```
ssh -Y taller-locmarcaXXX@kabre.cenat.ac.cr
```

Linux

Nothing to install: the X server is already there

In a terminal:

```
ssh -X taller-locmarcaXXX@kabre.cenat.ac.cr
```

Then the same steps: **echo $DISPLAY**, **which ncview**, **ncview file.nc &**

RECAP

Four commands to remember

```
ssh -X taller-locmarcaXXX@kabre.cenat.ac.cr # connect
echo $DISPLAY                               # not empty?
cd /data/gcambon/COMMONDATA/DATASETS_PSF    # go to the data
ncview COSTARICA_DS/sst_costarica_2023.nc & # look
```

ncview to look in seconds · **Python** to compute and make figures